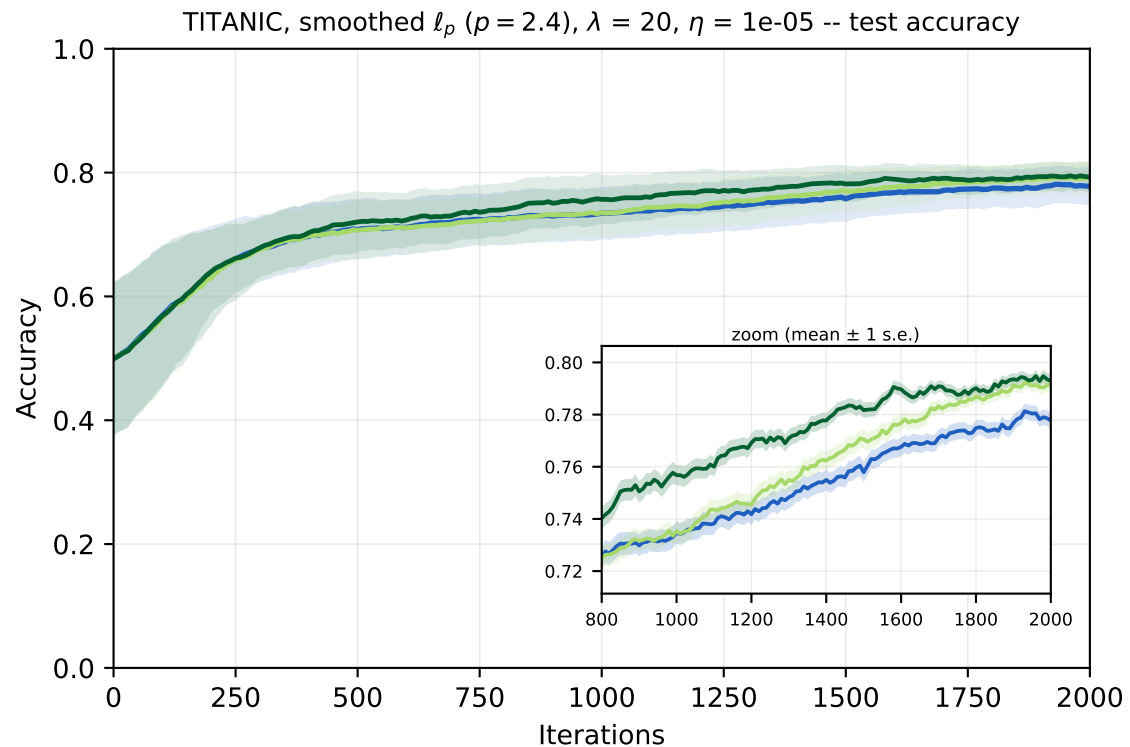
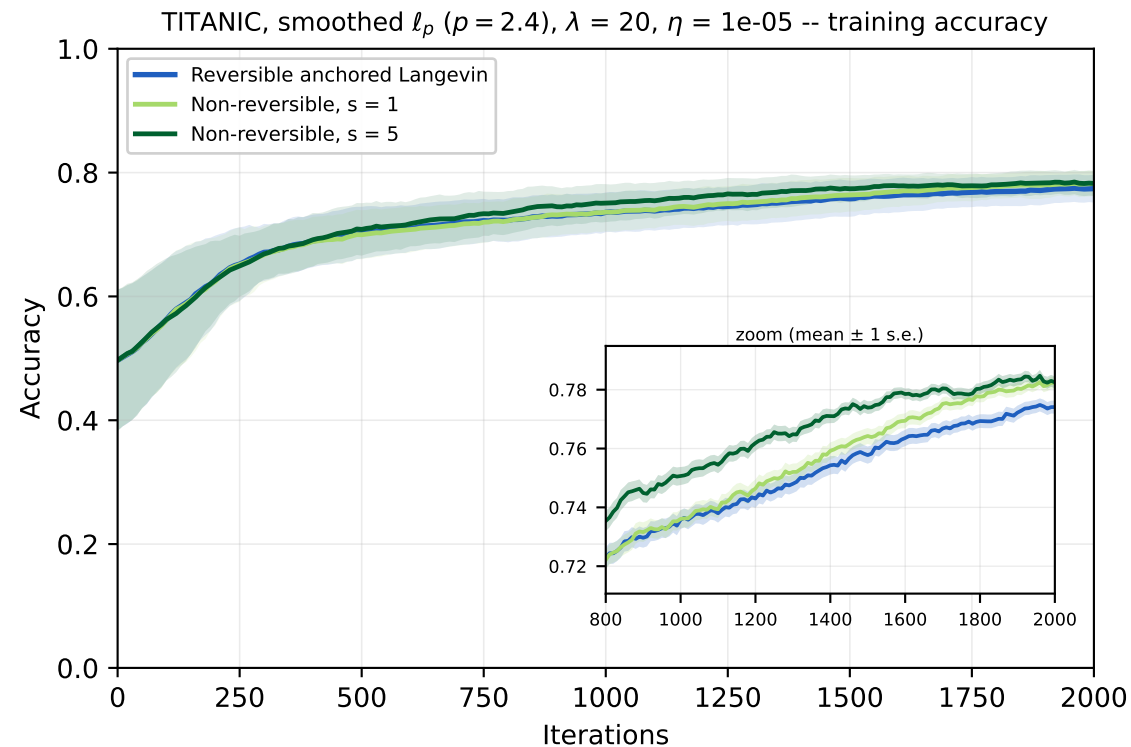


titanic_lp: $\lambda = 20$, $\eta = 1e-05$



TITANIC -- smoothed ℓ_p $g(w) = \sum_i (w_i^2 + \epsilon^2)^{p/2} \leq \text{Lambda}$, $p = 2.4$, $\epsilon = 0.18$, $\text{Lambda} = 4$. $d = 10$ (intercept $w_0 + 9$ features). $U = \sum \text{softplus} - y x \cdot w + w_0^2 / (2 \sigma^2) + \text{lambda} \sum_{j=1}^d |w_j|$; U_0 replaces $|w_j|$ by $\sqrt{w_j^2 + \delta^2}$; $a = \exp(U - U_0)$. EXACT gradient. $\sigma = 10$. $\text{lambda}_{\text{lasso}} = 20$ with $\delta = \log_2(9 \text{ lambda}) = 0.003851$, so a stays in $[0.50, 1]$ at every lambda (raising lambda also sharpens the smoothing -- the two are coupled by that choice). Step $\eta = 1e-05$. $R = 100$ independent replicates on one fixed stratified 80/20 split ($n_{\text{train}} = 712$, $n_{\text{test}} = 179$). Bands: mean ± 1 SD (ddof = 1) -- repeat-run variability at a fixed split, not a confidence or credible interval. All lambda are reported in results/lambda/lambda_lasso.csv.